Dynamics: Problem Sheet 8 (of 8)

1. A uniform cylinder of mass M, radius a and length l rolls, due to gravity $-g\mathbf{k}$, on the inside of a stationary larger cylinder of radius $A > a$. The symmetry axes of the two cylinders are parallel for all time and the centre of mass of the smaller cylinder, G, always remains in the **j**-**k** plane (see image).

   Let ψ denote the angle between the horizontal direction and the radial vector from the centre of the big circle to the centre of mass of the smaller cylinder, and let θ denote the change in angle of the smaller cylinder (from $t = 0$) as it rotates around its symmetry axis. Assuming the smaller cylinder rolls without slipping and contact between the cylinders is always maintained, you are given that

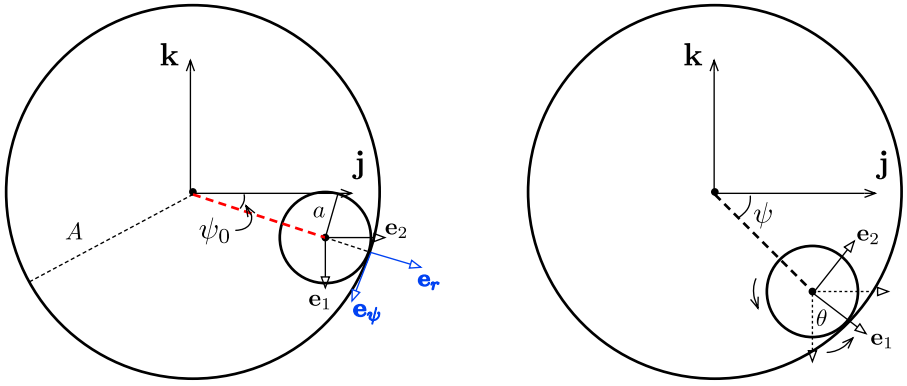
$$a\dot{\theta} = (A - a)\dot{\psi}.$$

   Determine the equation of motion for the small cylinder, in the form

$$\ddot{\psi} = \beta \cos\psi,$$

   where the constant β is to be determined.

   [*Hint: rolling without slipping implies the presence of a friction force* $-F\mathbf{e}_\psi$. *Write down the balances of linear and angular momentum – each will involve the unknown friction force; combine the two equations to eliminate* F.
   You may quote the inertia tensor about the centre of mass for the smaller cylinder rather than derive it.]



2. (a) A heavy pendulum consists of a uniform rigid rod of mass M andlength l, pivoted freely at one end, without loss the origin O, and swings freely in a vertical plane undergravity. Show that its kinetic energy is given by

$$T = \frac{1}{6}Ml^2\dot{\theta}^2 ,$$

   where θ is the angle the pendulum makes with the downward vertical.

   (b) Given that the potential energy is $V = MgZ_G$, where Z_G is the height of the centre of mass of the pendulum, hence write down the total energy $E = T + V$. Show that conservation of E is implied by the equation of motion (8.47) derived in the lecture notes.

3. A smooth straight wire rotates with constant angular speed ω about the vertical axis through a fixed point O on the wire, and the angle between the wire and the upward vertical is constant and equal to α, where $0 < \alpha < \pi/2$. A bead of mass m is free to slide on the wire.

   (a) Starting from the general form of Newton's second law in a rotating frame, show that

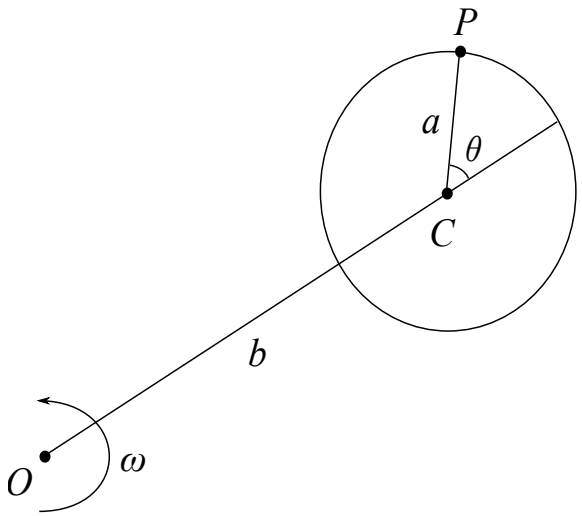
$$m\left(\frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2}\right)_{\mathcal{S}} = \mathbf{N} - mg\,\mathbf{k} - 2m\omega\,\mathbf{k}\wedge\left(\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}\right)_{\mathcal{S}} - m\omega^2\,\mathbf{k}\wedge(\mathbf{k}\wedge\mathbf{r})\ ,$$

   where $\mathbf{r}(t)$ is the position of the bead in the frame $\mathcal{S}$ that rotates with the wire, $\mathbf{k}$ is a unit vector pointing vertically, and $\mathbf{N}$ is the normal reaction of the wire.

   (b) Hence show that $z(t)$, the height of the bead above O, satisfies the equation

$$\ddot{z} - (\omega^2\sin^2\alpha)z = -g\cos^2\alpha\ .$$

   Show that an equilibrium point for the bead exists, and determine its stability.



4. A bead P of mass m slides on a smooth circular wire of radius a and centre C. The wire lies in a horizontal plane and is forced to rotate at a constant angular speed ω about the vertical axis through a fixed point O in the plane of the wire. The distance from O to C is constant and equal to b.

   Let θ be the angle that the line joining the centre C to the bead makes with the diameter through O. (See the figure below.)

   (a) Show that the position of the bead may be written as

$$\mathbf{r} = (b + a\,\cos\theta)\,\mathbf{e}_1 + a\,\sin\theta\,\mathbf{e}_2\ ,$$

   where $\mathbf{e}_i$, $i = 1, 2, 3$, are an orthonormal basis for a frame that rotates with the wire.

   (b) Using Newton's second law in this rotating frame, hence show that

$$\ddot{\theta} + \frac{b}{a}\omega^2\sin\theta = 0\ .$$

   (c) Show that the bead can remain in equilibrium relative to the wire at two points. Decide whether these positions of equilibrium are stable or unstable.